PolicyGraph - Complete Setup Guide

Prerequisites

- Python 3.9 or higher
- pip (Python package manager)
- Git

Step-by-Step Setup

1. Navigate to Repository

```
cd PolicyGraph  # Or wherever you cloned it
```

2. Create Virtual Environment

```
# Create the virtual environment
python3 -m venv .venv

# OR if python3 doesn't work, try:
python -m venv .venv
```

3. Activate Virtual Environment

On Linux/Mac:

```
source .venv/bin/activate
```

On Windows:

```
.venv\Scripts\activate
```

You should see `(.venv)` appear in your terminal prompt.

4. Upgrade pip

```
pip install --upgrade pip
```

5. Install PolicyGraph

```
# Install in editable mode with all dependencies
pip install -e .
```

This will install:
- torch (PyTorch)
- dgl (Deep Graph Library)

- scikit-learn
- PyYAML
- matplotlib
- and all other dependencies

6. Verify Installation

```
# Check version
python -c "import policygraph; print(f'PolicyGraph v{policygraph.__version__}')"

# Should print: PolicyGraph v0.1.0
```

7. Run Tests

```
# Install pytest if needed
pip install pytest

# Run all tests
pytest tests/ -v
```

8. Quick Test Run

```
# Analyze a sample policy
policygraph analyze data/raw/samples/escalation_passrole_lambda_create.json
```

Troubleshooting

Issue: "command not found: python3"

Solution: Try `python` instead of `python3`

Issue: "No module named 'torch'"

Solution: Run `pip install -e .` again

Issue: DGL installation fails

Solution:

```
# Install DGL manually (1.1.3 does NOT exist; use the 1.1.x range or pin 1.1.2)
pip install 'dgl>=1.1.0,<1.2.0' -f https://data.dgl.ai/wheels/repo.html
# Or pin to the latest 1.1.x release explicitly:
# pip install dgl==1.1.2 -f https://data.dgl.ai/wheels/repo.html
```

Issue: "policygraph: command not found"

Solution: Make sure virtual environment is activated and run `pip install -e .`

Quick Start Commands

After setup, you can use:

```
# Train model
policygraph train

# Evaluate model
policygraph evaluate

# Analyze single policy
policygraph analyze <policy.json>

# Batch analysis
policygraph batch data/raw/samples/

# Run baseline comparison
bash scripts/run_all_baselines.sh
```